BALROG: BENCHMARKING AGENTIC LLM AND VLM REASONING ON GAMES

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ABSTRACT

Recent progress in large language models has increased interest in evaluating models as interactive agents, but many current benchmarks still emphasize short-horizon tasks, low-interactivity settings, or narrowly scoped domains that are easy to saturate. We introduce BALROG, a benchmark for evaluating language-only and vision-language agents on long-horizon, procedurally generated game environments. BALROG unifies six lightweight environments, namely BabyAI, TextWorld, Crafter, Baba Is AI, MiniHack, and NetHack, under a common natural-language action interface, standardized prompting, and a shared 0–100 progression scale. The benchmark is designed to probe navigation, exploration, resource management, credit assignment, rule manipulation, and long-term planning across tasks whose human difficulty ranges from seconds to years. The released framework supports zero-shot evaluation, image-conditioned evaluation, in-context demonstrations, local serving, and public APIs, while logging invalid actions and per-episode trajectories for reproducibility. Baseline results show that current frontier models remain far from saturating the benchmark: language-only average progression peaks at 32.64%, vision-language performance is often worse than text-only performance, MiniHack remains difficult across model families, and NetHack is effectively unsolved, with the best model in the provided baselines still below 2% progression. These results position BALROG as a practical and diagnostic testbed for measuring long-horizon interactive reasoning rather than another static capability score.

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1 INTRODUCTION

Large language models (LLMs) and vision-language models (VLMs) are increasingly deployed as agents that must observe environments, choose actions, recover from mistakes, and plan over long horizons. This has created a demand for benchmarks that measure not only static reasoning, but also interactive competence under partial observability, long contexts, and non-trivial action consequences. Recent evaluation suites have expanded beyond static question answering to agentic settings, including interactive web tasks, tool use, and software engineering (Liu et al., 2023; Mialon et al., 2023; Yao et al., 2022; Zhou et al., 2023; Jimenez et al., 2023; Xing et al., 2024). However, many existing benchmarks still either involve relatively short trajectories, limited environmental diversity, or domains in which saturation can arrive before the underlying problems of exploration, planning, and sequential credit assignment are well understood.

Games remain a useful testbed for this problem because they are interactive, reproducible, and easy to instrument, while still supporting rich distributions over goals, maps, objects, and failure modes. In particular, procedurally generated games provide a strong substrate for measuring generalization beyond memorized instances. Prior work has used individual environments such as BabyAI (Chevalier-Boisvert et al., 2019), TextWorld (Côté et al., 2019), Crafter (Hafner, 2021), MiniHack (Samvelyan et al., 2021), and NetHack (Küttler et al., 2020; Goodger et al., 2023), as well as game-oriented benchmarks such as SmartPlay (Wu et al., 2023) and Baba Is AI (Cloos et al., 2024). What is still missing is a unified benchmark that spans easy and hard tasks, text-only and image-conditioned evaluation, and multiple kinds of agentic reasoning under a common protocol.

BALROG addresses this gap. It aggregates six environments into a single evaluation framework with a standardized interface, model-agnostic prompts, natural-language action spaces, and a consistent progression metric. The benchmark emphasizes long-horizon decision-making in procedurally gen-

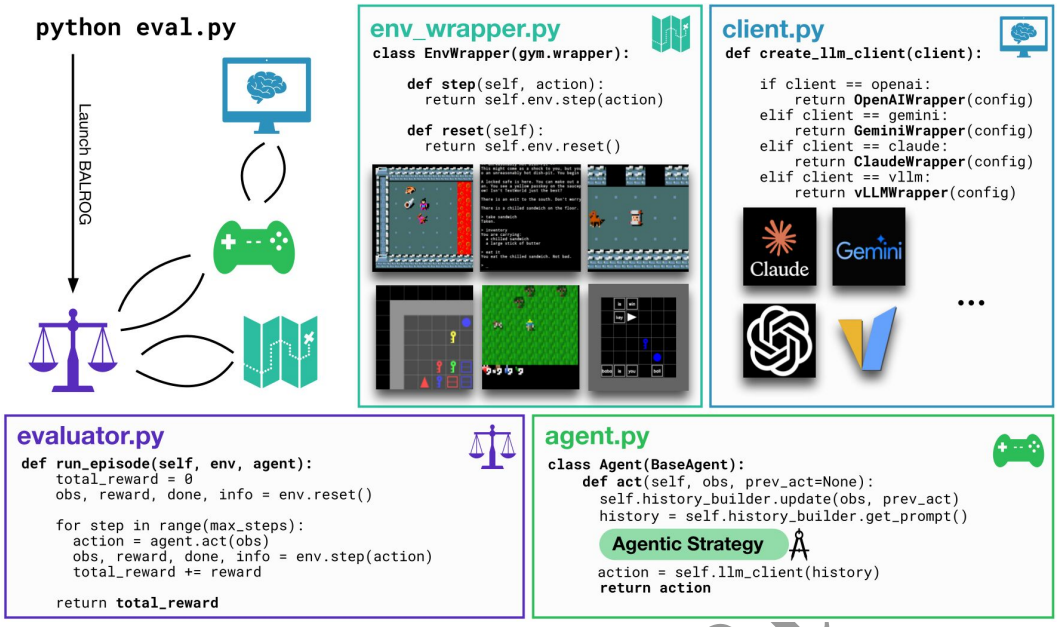

Figure 1: **Overview of BALROG.** A single evaluation entry point instantiates environment wrappers, model clients, and agent logic under a common interface. The framework supports multiple backends, exposes both text and image observations, and logs full trajectories for reproducible evaluation on long-context interactive tasks.

erated settings, from instruction following in BabyAI to open-ended exploration in NetHack. Rather than proposing a new learning algorithm, BALROG is intended to make evaluation more informative: it measures where current models succeed, where they fail, and how those failures differ across environments and modalities.

The benchmark contains 58 tasks across six environment families. It evaluates navigation, exploration, resource management, complex credit assignment, reasoning about environment dynamics, and long-term planning. These environments vary widely in horizon and difficulty, ranging from simple BabyAI tasks that skilled humans can solve in seconds to NetHack, a game with extreme horizon length and a long history of remaining challenging for both symbolic and learned agents (Küttler et al., 2020; Goodger et al., 2023). The released code additionally supports both API-based and locally served models, optional in-context demonstrations, and alternative prompting strategies, making BALROG useful both as a benchmark and as a controlled experimentation platform.

The headline empirical finding is sobering. The best language-only baseline in the provided artifacts, Claude 3.5 Sonnet, reaches only 32.64% average progression, narrowly ahead of GPT-4o at 32.34%. In the multimodal setting, Claude 3.5 Sonnet improves to 35.48%, but other models often degrade relative to their language-only counterparts. Per-environment results reveal that some tasks are partially tractable, especially BabyAI and Crafter, while MiniHack remains difficult and NetHack remains effectively unsolved, with the best plotted baseline still below 2% progression. These results suggest that current frontier models retain a substantial knowing–doing gap in interactive settings even when they perform strongly on static reasoning benchmarks.

Contributions. This paper makes four contributions:

- It introduces BALROG, a unified benchmark for evaluating LLM and VLM agents on six procedurally generated game environments under a shared action and scoring interface.
- It provides a reproducible evaluation protocol with standardized prompting, history management, invalid-action handling, and support for both closed and open model backends.

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- It defines a common 0–100 progression view across heterogeneous tasks, including binary success settings, proportional achievement metrics, and a non-linear NetHack progression statistic implemented through a fixed lookup over gameplay milestones.
- It reports zero-shot baselines that reveal persistent weaknesses in long-horizon planning, exploration, and visual decision-making for current frontier models.

2 Related Work

2.1 Benchmarks for LLM Agents

Interactive evaluation has become a central concern as language models are increasingly used as agents. AgentBench (Liu et al., 2023), SmartPlay (Wu et al., 2023), WebShop (Yao et al., 2022), WebArena (Zhou et al., 2023), GAIA (Mialon et al., 2023), SWE-bench (Jimenez et al., 2023), and AndroidArena (Xing et al., 2024) all expose important dimensions of agent behavior, such as tool use, web navigation, software editing, and long-horizon task completion. These benchmarks have shown that strong static language models remain brittle when embedded in sequential decision problems. BALROG is complementary to this line of work. Its main difference is a focus on procedurally generated game environments with lightweight simulation, standardized natural-language actions, and a wider span of horizon lengths within a single benchmark. This makes it possible to compare models across easy and hard interactive tasks without relying on a single application domain.

2.2 Game-Based and Environment-Centric Evaluation

Several of the constituent environments in BALROG have already supported important research agendas. BabyAI studies grounded instruction following (Chevalier-Boisvert et al., 2019). TextWorld provides generated text games for sequential language grounding (Côté et al., 2019). Crafter measures open-ended exploration and achievement completion (Hafner, 2021). MiniHack and NetHack are challenging RL platforms with rich dynamics and long horizons (Samvelyan et al., 2021; Küttler et al., 2020). Baba Is AI focuses on rule manipulation and puzzle solving (Cloos et al., 2024). Prior work has also explored language agents directly in some of these environments, for example through NetHack language wrappers and history compression (Goodger et al., 2023; Piterbarg et al., 2024). BALROG does not replace these individual environments. Instead, it integrates them into a single benchmark with common prompting, action handling, and scoring so that cross-environment comparisons become practical.

2.3 Long-Context and Multimodal Evaluation

BALROG is also related to recent efforts to study long-context decision-making and multimodal control. LMAct evaluates in-context imitation learning with long multimodal demonstrations (Ruoss et al., 2024), while recent VLM diagnostics have highlighted persistent failures in visual grounding and spatial reasoning (Rahmanzadehgervi et al., 2024). These observations motivate the design of BALROG's paired language-only and vision-language settings. Unlike single-task multimodal evaluations, BALROG measures whether adding images improves decision-making across interactive tasks with different visual and planning requirements.

3 Benchmark Design

3.1 Design Goals

The benchmark was designed around five requirements. First, it should test interactive reasoning rather than static recall. Second, it should span short and very long horizons. Third, it should resist trivial memorization through procedural variation whenever possible. Fourth, it should support both text-only and multimodal agents under a common interface. Fifth, it should be easy to reproduce and extend in practice.

These goals led to an environment set composed of lightweight RL-style games rather than expensive embodied simulators or static datasets. The selected tasks differ substantially in mechanics

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Table 1: **Capabilities and difficulty spectrum covered by BALROG.** The benchmark spans a wide range of interactive skills and horizon lengths, from simple instruction following to open-ended long-horizon exploration.

Skill / Property	**BabyAI**	**TextWorld**	**Crafter**	**Baba Is AI**	**MiniHack**	**NetHack**
Navigation	✔	✔	✔	✔	✔	✔
Exploration	✔	✔	✔	✔	✔	✔
Resource management	✗	✔	✔	✗	✔	✔
Complex credit assignment	✗	✗	✔	✔	✔	✔
Deducing environment dynamics	✗	✗	✗	✔	✔	✔
Long-term planning	✗	✗	✗	✔	✔	✔
Typical turns to complete	10^1	10^2	10^3	10^2	10^2	10^4–10^5
Time to master for humans	Seconds	Minutes	Hours	Hours	Hours	Years

while still permitting unified evaluation through text observations, natural-language actions, and per-episode progression scores.

3.2 Environment Suite

Table 1 summarizes the capabilities stressed by each environment. BALROG includes five BabyAI instruction-following tasks, three TextWorld tasks, one Crafter task, 40 Baba Is AI levels, eight MiniHack tasks, and one NetHack Challenge task. The environments cover:

- **BabyAI** (Chevalier-Boisvert et al., 2019): short-horizon instruction following and navigation.
- **TextWorld** (Côté et al., 2019): text-only exploration and sequential interaction with sparse task structure.
- **Crafter** (Hafner, 2021): open-ended resource gathering, tool construction, and achievement progression.
- **Baba Is AI** (Cloos et al., 2024): puzzle solving that requires manipulating rules as well as objects.
- **MiniHack** (Samvelyan et al., 2021): navigation, combat, and puzzle-style NetHack derivatives.
- **NetHack** (Küttler et al., 2020; Goodger et al., 2023): extremely long-horizon, high-branching gameplay with rich partial observability and delayed rewards.

3.3 Unified Observation and Action Interface

All environments are wrapped behind a common interface that exposes a reset function, a step function, environment-specific instruction prompts, a language action space, and summary statistics. Actions are represented in natural language. For example, agents emit strings such as `go forward` in BabyAI, `Move North` in Crafter, `open` or `search` in MiniHack, and free-form text commands in TextWorld. This design makes it possible to evaluate models without task-specific decoders or learned action heads.

Observations are standardized into a text channel and, when available, an image channel. The text channel is split into `long_term_context` and `short_term_context`, which are then serialized into a dialogue-style prompt history. In text-only mode, the agent sees only textual observations. In multimodal mode, the framework additionally attaches rendered images to recent observations. The precise textual content is environment dependent. For example, NetHack and MiniHack provide textual state descriptions and, in the language-only condition, an ASCII map; the multimodal condition replaces that map with rendered tiles while retaining textual status information. This pairing allows the benchmark to test whether image access helps action selection rather than only visual recognition.

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3.4 Prompting and History

The default baseline agent is deliberately simple. It receives a task-specific instruction prompt, the recent observation-action history, and a final instruction that it must output exactly one valid action at each turn. The released configuration keeps the last 16 text interactions and no images for language-only runs. The same infrastructure supports image history for VLM runs, chain-of-thought-style agents, and in-context demonstrations, but the baselines in the provided artifacts use zero-shot prompting.

This design choice matters for interpretation. The benchmark measures what strong pretrained models can do with a standardized lightweight agent shell, not the best achievable performance after prompt engineering or search. Accordingly, the baseline numbers should be read as reference points rather than ceilings.

3.5 Scoring

BALROG maps each episode to a progression score in $[0, 1]$, reported as a percentage. The scoring rule depends on the environment:

- **BabyAI, Baba Is AI, and MiniHack**: binary success, scored as 1 for successful completion and 0 otherwise.
- **Crafter**: fraction of the 22 achievements unlocked in the episode.
- **TextWorld**: normalized terminal game score, with a floor at binary win/loss.
- **NetHack**: a fixed non-linear progression value based on the highest dungeon depth or experience level reached during the episode.

The NetHack metric is especially important because sparse raw return makes direct comparison difficult. The implementation tracks the maximum achieved milestone and converts it to progression through a lookup table over dungeon levels and experience levels, with deeper and higher-level milestones receiving larger values. This produces a dense scalar summary that still respects the structure of progress in the game.

For a given model, the overall benchmark score is the unweighted mean of the six environment-level progression means computed by the evaluation code. This choice prevents environments with many tasks or episodes from dominating the headline number.

4 Evaluation Protocol

4.1 Tasks, Episodes, and Seeds

The released default configuration evaluates all six environment families with multiple episodes per task. Specifically, the repository configuration runs 10 episodes for each BabyAI task, 10 for each TextWorld task, 10 for Crafter, 3 for each Baba Is AI level, 5 for each MiniHack task, and 5 for NetHack, for a total of 255 episodes per model across the full benchmark. When no fixed seed is supplied, the evaluator samples episode seeds automatically, which is important for procedurally generated tasks.

4.2 Invalid Actions and Feedback

A benchmark intended for broad model coverage must handle malformed outputs robustly. BALROG therefore validates each model output against the environment's language action space. If the output is invalid, the evaluator logs the failed candidate, executes an environment-specific fallback action, and can prepend feedback about the invalid action to the next observation. This policy makes evaluation resilient to formatting mistakes while preserving information about model reliability.

4.3 Infrastructure and Reproducibility

The framework supports multiple client backends, including OpenAI, Anthropic, Gemini, and locally served models through vLLM (Kwon et al., 2023). Evaluations can be parallelized across

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workers, resumed from partial runs, and configured through Hydra. The code also logs per-step trajectories, token usage, action frequencies, failed action candidates, and environment-specific summary statistics. Because the benchmark is built on existing simulators rather than human annotation, there is no annotation stage or inter-annotator agreement analysis. Reproducibility instead depends on fixed configurations, public code, logging, and the procedural generation properties of the underlying environments.

4.4 Models

The provided artifacts include zero-shot baselines for a mix of closed and open models. Language-only results are reported for Claude 3.5 Sonnet (Anthropic, 2024), GPT-4o (OpenAI, 2024a), Gemini 1.5 Pro and Flash (Reid et al., 2024), and several Llama 3.1/3.2 instruction-tuned models (Dubey et al., 2024; MetaAI, 2024). The multimodal setting includes the major vision-capable subsets of these families. The per-environment baseline figure additionally includes o1-preview (OpenAI, 2024b) for NetHack.

5 Benchmark Validation

5.1 Coverage and Diversity

BALROG is not a single game with many seeds; it is a cross-environment benchmark that intentionally spans qualitatively different interactive demands. BabyAI emphasizes short-horizon instruction execution, TextWorld tests text-based exploration and command composition, Crafter emphasizes open-ended progression, Baba Is AI tests rule manipulation, MiniHack adds compact but difficult NetHack-style tasks, and NetHack probes long-range exploration and planning at an extreme horizon. This diversity is a core validity property: a benchmark meant to assess agentic reasoning should not collapse onto a single mechanic.

5.2 Contamination Resistance

Procedural generation improves contamination resistance by reducing the chance that memorizing fixed instances suffices for strong performance. This is strongest in NetHack, MiniHack, Crafter, and the generated TextWorld games, and it is weaker where the task families are small or templates are more structured. We therefore view BALROG as more resistant to memorization than static benchmark sets, but not as immune to contamination in the absolute sense.

5.3 Metric Sanity Checks

The benchmark's scoring rules are intentionally simple and inspectable. Binary completion is used when success is unambiguous, achievement fraction is used when progression is naturally cumulative, and the NetHack statistic is exposed directly through a fixed milestone table. The resulting baseline spread indicates that the benchmark is neither trivial nor degenerate. There is measurable headroom at the easy end of the suite, clear separation among models on intermediate tasks, and essentially no saturation on the hardest tasks. This combination is desirable for a benchmark that aims to remain informative over time.

5.4 Known Scope Boundaries

BALROG evaluates interactive decision-making under constrained interfaces, not general intelligence in the abstract. It does not directly measure social reasoning, web tool use, embodied robotics, or safety-critical deployment behavior. It also does not attempt to identify the optimal prompting strategy for each model. The released baselines are intentionally standardized and lightweight so that differences in performance are easier to attribute to model capabilities rather than benchmark-specific prompt tuning.

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Table 2: **Aggregate BALROG results.** Scores are average progression percentages as reported by the released evaluation code. The overall score is the unweighted mean of environment-level progression means. Best results in each column are bolded.

Model	Language-only	Vision-language
claude-3.5-sonnet	**32.64 ± 1.93**	**35.48 ± 2.02**
gpt-4o	32.34 ± 1.49	22.56 ± 1.44
llama-3.1-70b-it	27.88 ± 1.43	–
llama-3.2-90B-it	27.29 ± 1.44	20.99 ± 1.58
gemini-1.5-pro	21.00 ± 1.18	25.76 ± 1.36
claude-3.5-haiku	19.32 ± 1.83	–
gpt-4o-mini	17.36 ± 1.35	15.36 ± 1.29
llama-3.2-11B-it	16.82 ± 1.47	8.43 ± 1.26
llama-3.1-8b-it	15.14 ± 1.55	–
gemini-1.5-flash	14.63 ± 1.37	14.94 ± 1.40
llama-3.2-3B-it	10.13 ± 1.28	–
llama-3.2-1B-it	6.65 ± 1.04	–

6 Benchmark Results

6.1 Overall Performance

Table 2 reports the aggregate results provided in the benchmark artifacts. The best language-only model is Claude 3.5 Sonnet at 32.64% average progression, closely followed by GPT-4o at 32.34%. Among open models, Llama 3.1-70B reaches 27.88%, while smaller Llama variants drop substantially. In the vision-language setting, Claude 3.5 Sonnet improves to 35.48%, Gemini 1.5 Pro reaches 25.76%, and GPT-4o drops to 22.56%.

Two patterns stand out. First, all absolute scores are modest. Even the strongest models complete only about one third of the benchmark on average. Second, multimodal access is not uniformly helpful. Claude 3.5 Sonnet benefits from images, but GPT-4o and the Llama 3.2 vision model both perform substantially worse in the vision-language setting than in text-only evaluation. This suggests that access to pixels does not automatically translate into better control, and in some cases may instead create additional perception-action failure modes.

6.2 Per-Environment Trends

Figure 2 shows the per-environment baselines. BabyAI is the most tractable environment, with several models reaching high partial success. Crafter yields moderate progress across many model families, indicating that cumulative achievement-style environments are useful for separating models without collapsing to all-or-nothing outcomes. TextWorld is brittle: some models achieve strong progress, while others are near zero. Baba Is AI remains clearly challenging, consistent with the benchmark's emphasis on changing rules rather than only following them. MiniHack is difficult across the board, with progress typically in the low tens or below. NetHack is the hardest environment by a wide margin; the best plotted model remains below 2% progression.

7 Analysis

7.1 BALROG Separates Short- and Long-Horizon Competence

The environment breakdown shows that strong performance on simpler or shorter-horizon tasks does not transfer cleanly to harder settings. Several frontier models achieve non-trivial BabyAI, Crafter, or TextWorld scores, yet the same models collapse on NetHack and struggle on MiniHack. This is precisely the kind of separation a useful benchmark should expose. A model that appears broadly capable on short interactive tasks may still lack the exploration, state tracking, or long-term planning required for extended control.

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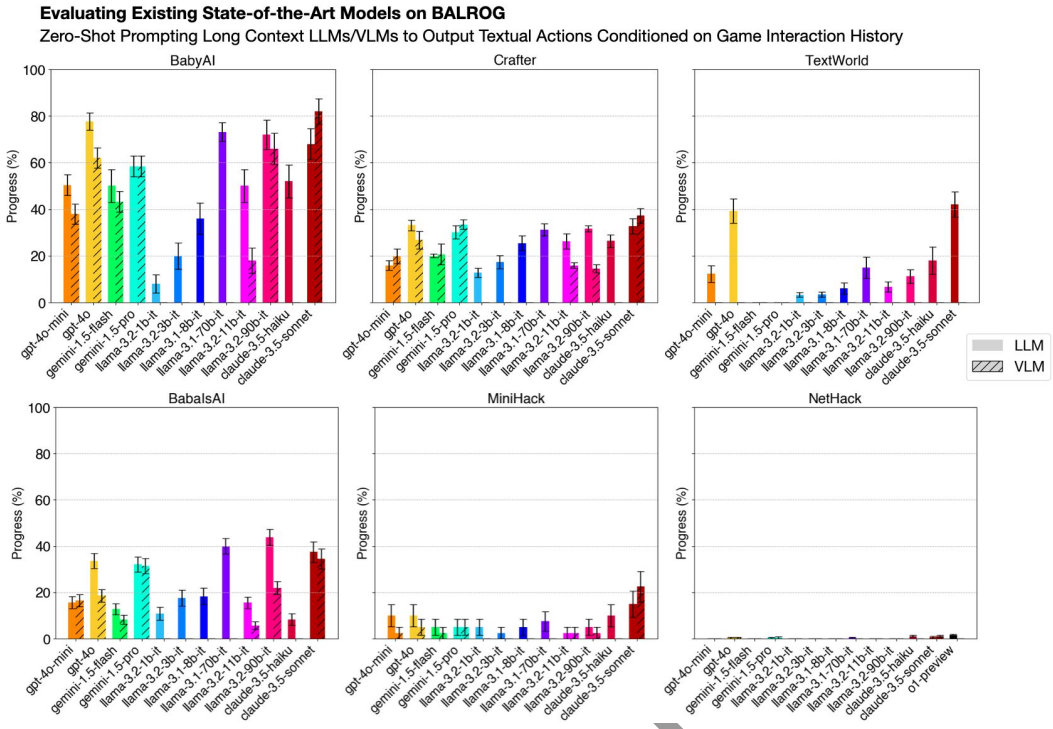

Figure 2: **Per-environment baselines for BALROG.** The plot reveals substantial heterogeneity across environments. BabyAI and Crafter provide partial traction, Baba Is AI and MiniHack remain difficult, and NetHack is essentially unsolved in the released baselines. Hatched bars denote vision-language evaluation.

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7.2 Vision Often Hurts Control

One of the clearest empirical findings is that multimodal inputs do not reliably improve performance. This is noteworthy because the evaluated VLMs are designed to accept image inputs, and several are state of the art on conventional multimodal benchmarks. On BALROG, however, image conditioning often makes performance worse. This aligns with recent evidence that VLMs can still struggle with precise visual grounding and spatial reasoning even when they perform well on higher-level recognition tasks (Rahmanzadehgervi et al., 2024). In interactive games, those weaknesses are amplified because perception errors translate directly into bad actions.

7.3 The Benchmark Exposes a Knowing–Doing Gap

Many of the evaluated environments are simple to explain in language. The rules of BabyAI, Crafter, and several MiniHack tasks are explicit in the prompts, and the allowed action sets are enumerated. Nevertheless, models still fail frequently. This suggests that the main challenge is not merely understanding the instructions, but converting that understanding into robust sequential behavior. Such a gap has been discussed in broader critiques of LLM planning and grounded control (Carta et al., 2023; Kambhampati et al., 2024). BALROG makes that gap observable across multiple interactive domains within one benchmark.

7.4 Why NetHack Matters

NetHack deserves special attention because it serves as the benchmark's long-horizon extreme. The environment combines large action spaces, partial observability, inventory management, delayed rewards, and long dependency chains (Küttler et al., 2020). The near-zero baseline scores in Figure 2 show that current general-purpose LLMs and VLMs are still far from robust competence in this

regime. This is useful scientifically. A benchmark should contain tasks that remain unsolved and diagnostic rather than only tasks that produce leaderboard churn in the saturated regime.

8 Discussion

BALROG is best viewed as an evaluation framework for interactive reasoning, not as a claim that one benchmark can summarize all of agency. Its main contribution is methodological: it standardizes a diverse set of procedurally generated environments under one protocol and shows that this protocol yields informative, non-saturated, and reproducible measurements.

Several limitations are important. First, the current paper reports only standardized zero-shot baselines; stronger prompting, search, or in-context strategies may improve absolute performance. Second, because the benchmark reuses existing simulators, the environments vary in observation format, reward structure, and interface complexity. A shared progression scale helps, but it does not erase these differences. Third, no human baseline is included in the provided artifacts, so claims about human-relative difficulty should be interpreted through the original environment literature and the coarse difficulty summary in Table 1, not through direct human experiments here. Fourth, multimodal evaluation is mediated by current API implementations and benchmark wrappers, so part of the observed VLM degradation may arise from the full perception-to-action stack rather than vision alone.

Interpreting aggregate scores. BALROG should be read as a capability profile rather than only a leaderboard scalar. Similar overall means can mask sharply different behavior across TextWorld, Crafter, Baba Is AI, and MiniHack, and language-only versus vision-language evaluation can also reverse rankings.

- report the aggregate mean together with per-environment scores;
- treat text-only and vision-language comparisons as separate views rather than interchangeable summaries.

These limitations do not reduce the value of the benchmark; they define how it should be used. BALROG is most informative when used to compare agent designs, prompting strategies, or model families under a fixed protocol, and when paired with per-environment analysis rather than only a single scalar score. Because the toolkit also supports few-shot demonstrations and alternative agents, it can serve as a common testbed for studying which interventions genuinely improve long-horizon interactive behavior.

9 Conclusion

We presented BALROG, a benchmark for evaluating LLM and VLM agents on six procedurally generated game environments spanning short- and long-horizon interactive reasoning. The benchmark unifies environment interfaces, natural-language actions, prompting, and progression scoring while remaining practical to run and reproduce. The provided baselines show that current models make partial progress on easier environments, but remain weak on harder sequential control problems; multimodal inputs often fail to help, and NetHack remains essentially unsolved. These findings suggest that long-horizon interactive evaluation is still necessary even for models that appear strong on static benchmarks, and that BALROG provides a useful measurement tool for tracking that progress. More broadly, the benchmark is valuable because it distinguishes different failure modes instead of collapsing them into one game-specific score, making it easier to tell whether an apparent gain reflects better exploration, better control, or simply an easier slice. In practice, this makes slice-level reporting a requirement rather than an optional supplement.

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